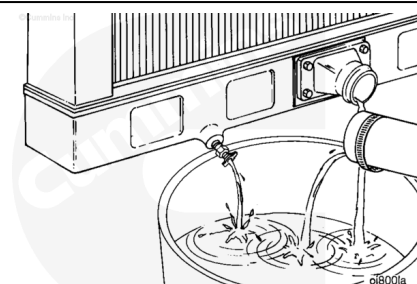


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

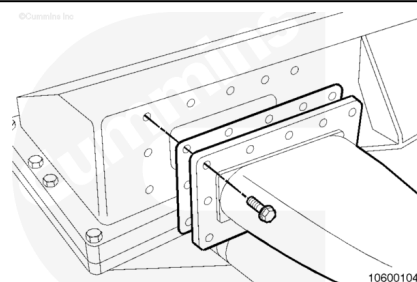
Remove

CELECT™ or Centry™

Note : The front intercooler element removal procedure is shown. The rear intercooler element removal procedure is similar.

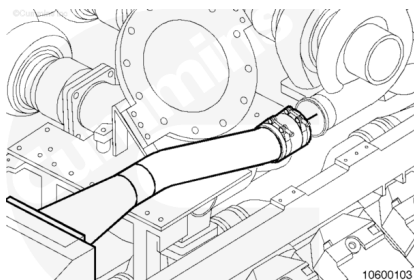
Remove the 12 capscrews each from four intercooler intake crossovers and the two intercoolers.

Remove and discard the gasket.

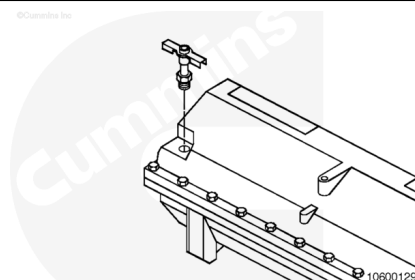


Loosen the hose clamps on each end of the four air connection hoses for the four intercooler intake crossovers and the four low-pressure turbochargers.

Remove the four intercooler intake crossovers.

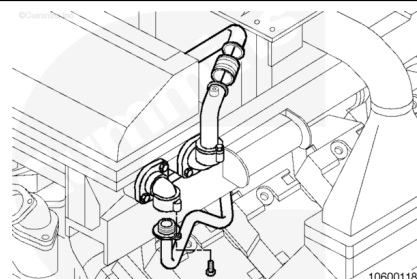


Remove the petcock vent from the intercooler.

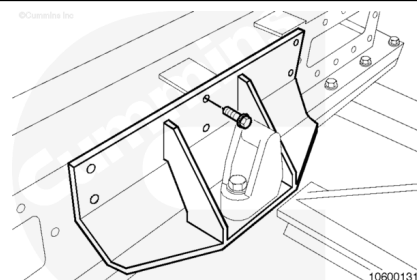


Note : The front intercooler water supply and return connections are shown. The rear intercooler connections are similar.

Disconnect the water supply and return piping from the intercoolers.



Remove six capscrews from the bracket and intercooler.

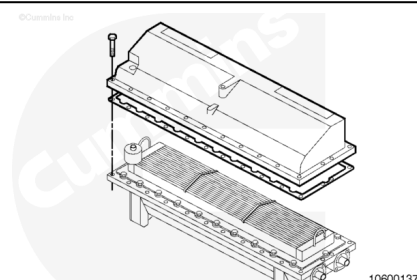


⚠ WARNING ⚠

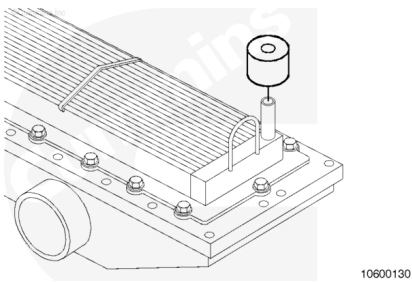
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to remove this assembly.

Install a 12-mm [0.47-in] lifting device in the top of the intercooler.

Remove the 26 capscrews and the intercooler cover and gasket from the intercooler.

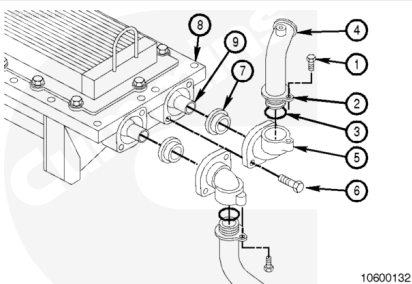


Remove the air grommet from the air vent tube.



Remove one capscrew (1), flange (2), seal (3), and pipe (4) each from two intercooler fittings (5).

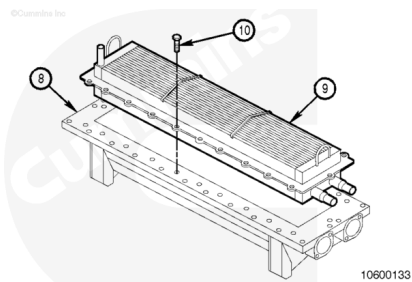
Remove four capscrews (6), two intercooler fittings (5), and seals (7) from the intercooler (8) and element (9).



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to remove this assembly.

Remove the 22 capscrews (10) and the element (9) from the intercooler (8).



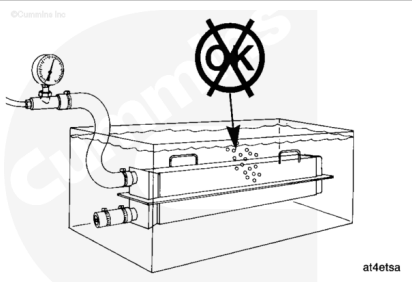
Pressure Test

Heating the tank water to 50°C [122°F] improves the test results.

Use a hose, plugged with a pipe plug, to plug one water tube as shown.

Attach an air line and gauge to the remaining water tube.

Apply air pressure, submerge the intercooler element, and check for leaks.



Measurements

	kpa	psi
Air Pressure:	415	60

Do **not** attempt to repair a leaking element. If an element has a leak, it **must** be replaced.

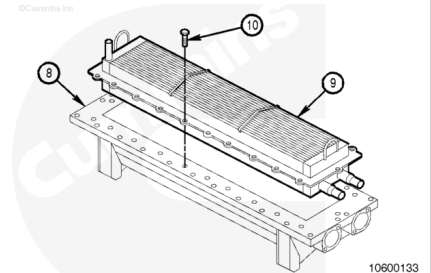
Install

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to remove this assembly.

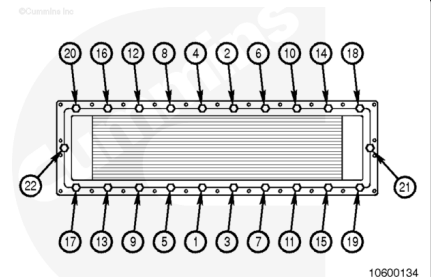
The intercooler element **must** be positioned on the intercooler alignment dowels during installation.

Position the element (9) in the lower intercooler housing (8) with 22 capscrews (10).

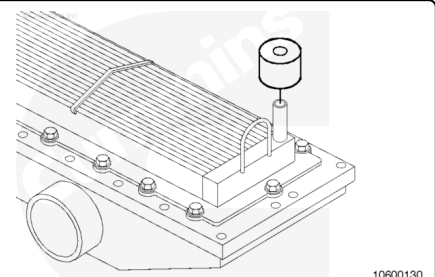


Tighten the capscrews in the pattern shown.

Torque Value: 45 n·m [33 ft·lb]



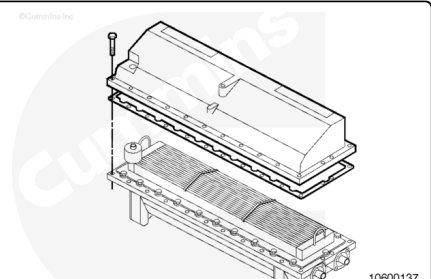
Install the air grommet on the air vent tube.



⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to remove this assembly.

Install a 12-mm [0.47-in] lifting device in the top of the intercooler cover.

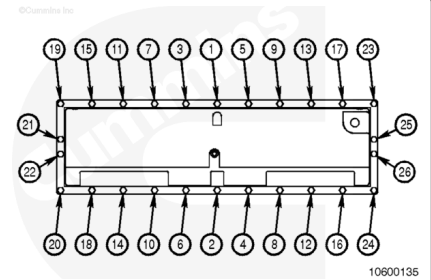


Slightly tighten the four capscrews near the air grommet to hold the air grommet in position.

Position the intercooler gasket and cover on the intercooler with 26 capscrews.

Tighten the capscrews in the pattern shown.

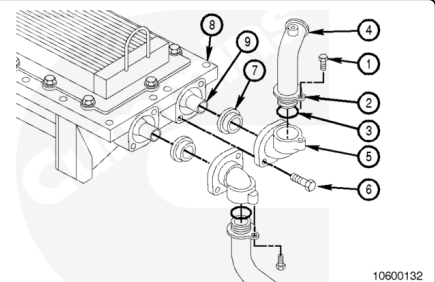
Torque Value: 45 n•m [33 ft-lb]



Install the flange (2), seal (3), and pipe (4) in each of two intercooler fittings (5) with one capscrew (1).

Install the two intercooler fittings (5) and seals (7) in the intercooler (8) and element (9) with four capscrews (6).

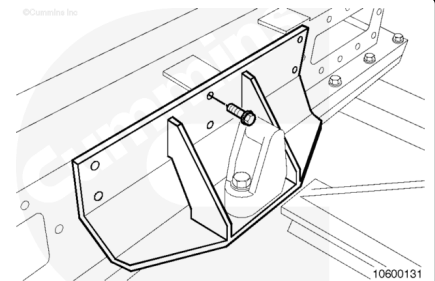
Torque Value: 45 n•m [33 ft-lb]



Install the bracket to the intercooler with six capscrews.

Tighten the capscrews.

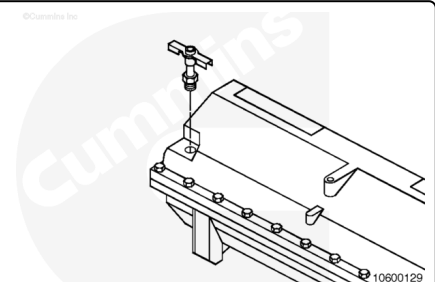
Torque Value: 45 n•m [33 ft-lb]



Apply pipe sealant to the threads of the petcock.

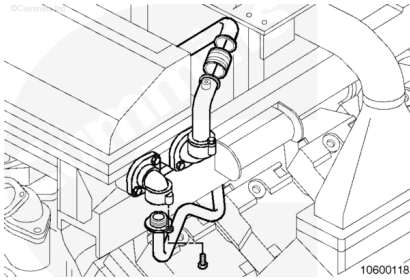
Install the petcock vent on the intercooler.

Torque Value: 8 n•m [71 in-lb]



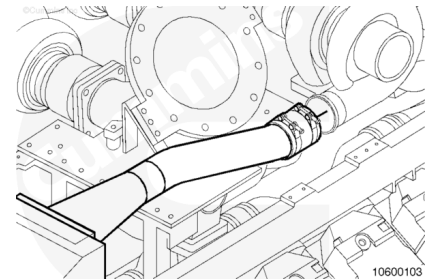
Note : The front intercooler water supply and return connections are shown. The rear intercooler connections are similar.

Connect the water supply and return piping to the intercoolers.



Install the four air connection hoses on the four intercooler intake crossovers and the four low-pressure turbochargers with two clamps each.

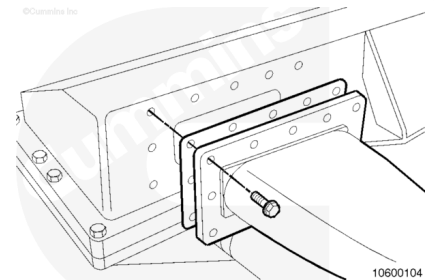
Torque Value: 9 n•m [80 in-lb]



Note : The front intercooler element installation procedure is shown. The rear intercooler element installation procedure is similar.

Install the four intercooler intake crossovers and gaskets on the two intercoolers with 12 capscrews each.

Torque Value: 45 n•m [33 ft-lb]



Finishing Steps

Fill the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

